

▷ UNLIKELY PEAKS

There has been enough vertical movement along the usually horizontal fault that cuts through New Zealand to throw up the Southern Alps.



△ UNIQUE WILDLIFE

As a result of Australia's isolation from other continents, much of its wildlife has evolved independently and is unique to the continent.

away, too, and headed north—but Australia was hauled to the chilly South Pole in Antarctica's embrace. From 85 million years ago, Australia began to crack away from Antarctica until, about 45 million years ago, it finally broke free and started creeping northward—and it is still heading in that direction.

Although the Australian Plate is now pressing up against the Eurasian Plate, unlike the Indian Plate it has never come into direct contact by land—Australia and the nearby islands remain separated by deep water. Australia's wildlife has evolved independently, and some species are completely unique to the continent.

Hidden continent

The landmass that was to become New Zealand split away from Antarctica and Australia 80 million years ago. A great rip formed along Australia's western edge and along the edge of Marie Byrd Land in Antarctica. New Zealand drifted north, and the Tasman Sea opened to leave the two islands isolated.

In 2017, scientists confirmed what had long been suspected—that the split moved not just present-day New Zealand but an entire continent, over 90 percent of which is now submerged.

This submerged continent, known as Zealandia, is now considered the world's eighth continent due to the type of crustal rock that it is made of. New Zealand is, in fact, half on the Australian Plate and half on the Pacific Plate, and for 25 million years has been torn apart along the Alpine Fault (which has thrown up the Southern Alps) as the two plates move in opposite directions.

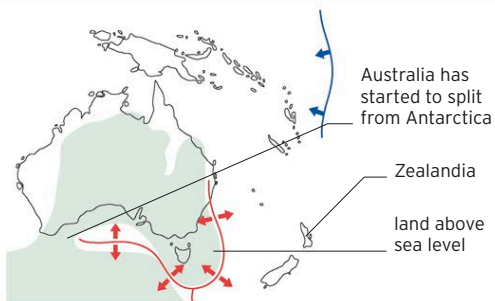
HIDDEN CONTINENT

The islands of New Zealand and New Caledonia, although now separated by sea, are just the highest points of the now submerged continent of Zealandia.

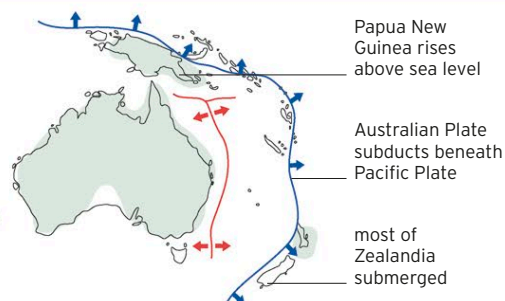


BUILDING THE AUSTRALIAN AND ZEALANDIAN CONTINENTS

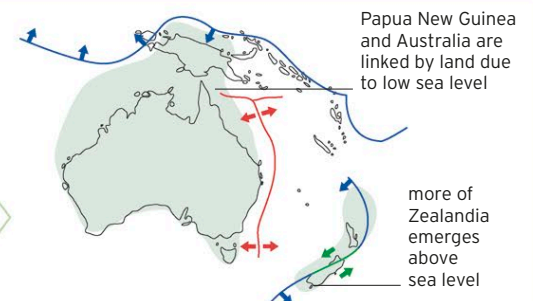
KEY — Convergent boundary — Divergent boundary — Transform boundary



94 MILLION YEARS AGO Australia and Zealandia are still attached to Antarctica as part of the same continental fragment of Gondwana, but the rift between them all will soon start to open.



50-40 MILLION YEARS AGO Australia and Zealandia are now separate from Antarctica. Australia is gliding quickly north toward the tropics, while Zealandia is moving northeast, largely underwater.



18,000 YEARS AGO Australia is close to the south of Asia. The drop in sea levels during the ice age causes parts of Zealandia to emerge and Papua New Guinea to connect to Australia.